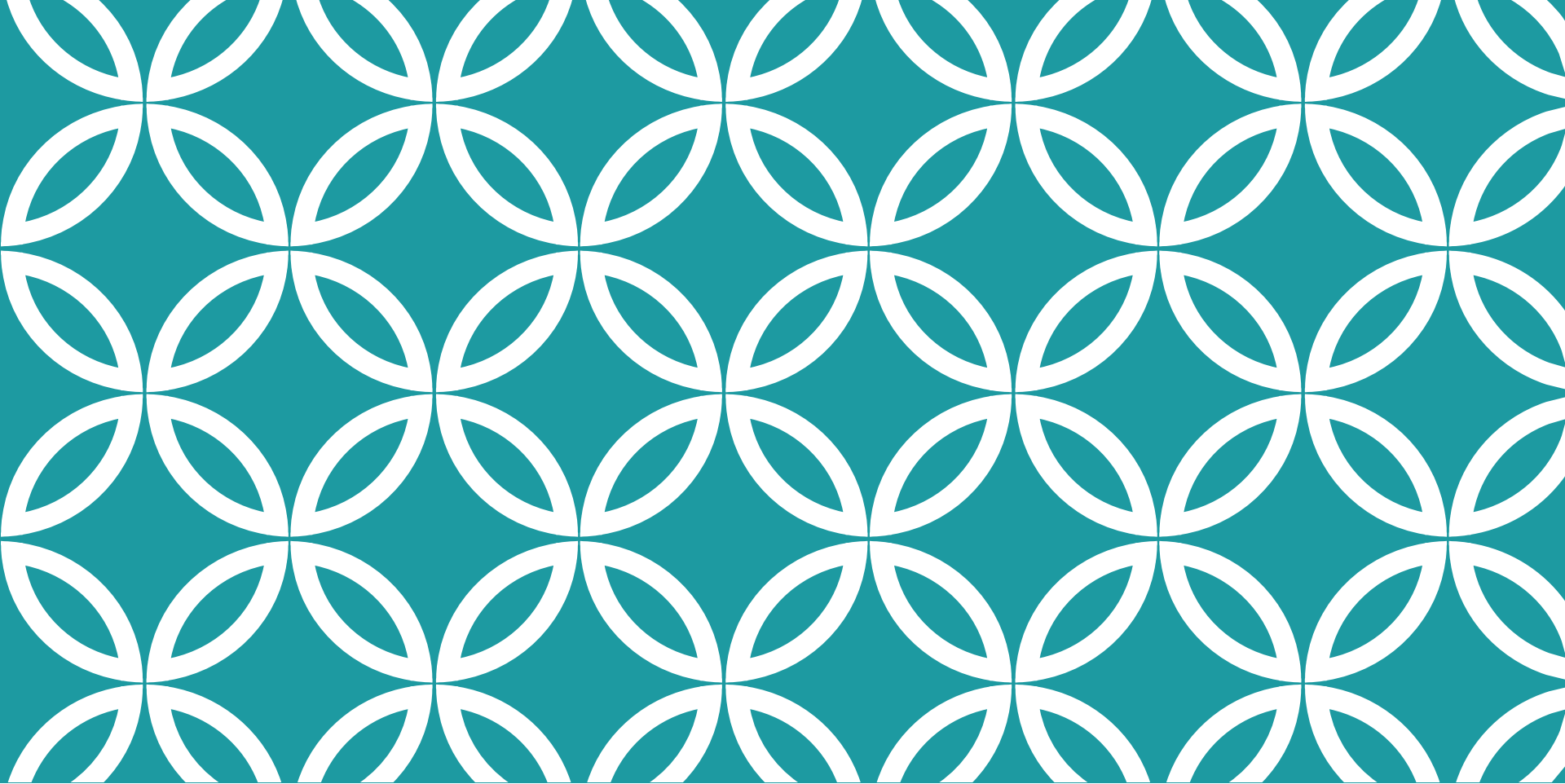


# SPEECH SIGNAL REPRESENTATION

## Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.



# FREQUENCY-DOMAIN N REPRESENTATIONS

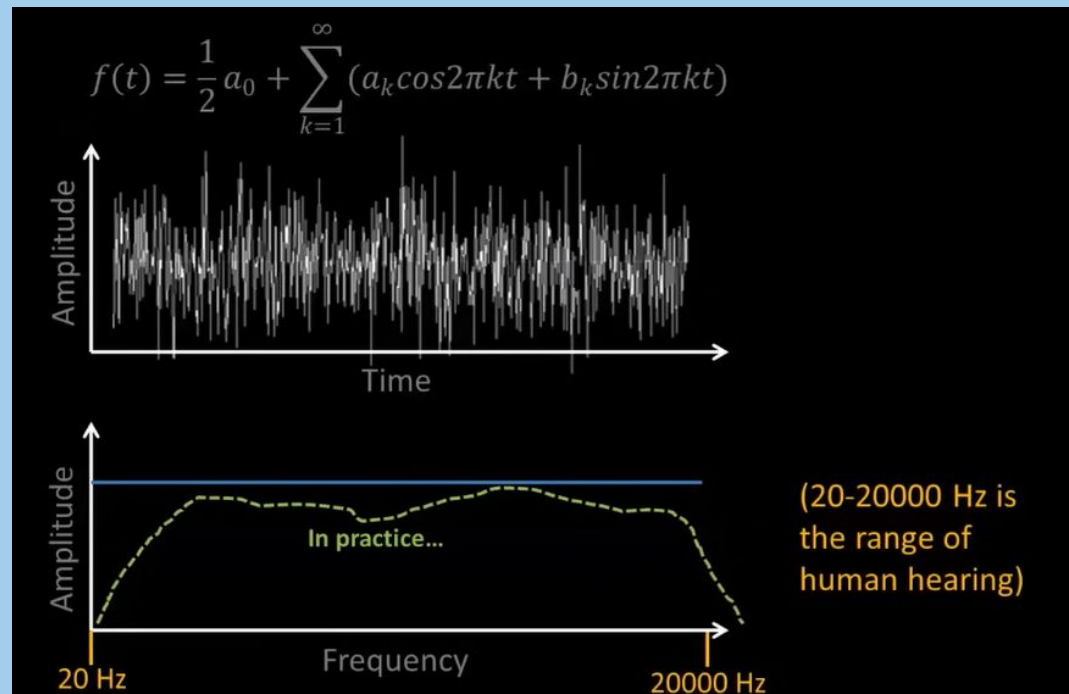
Why time-domain to  
frequency-domain  
Fourier Transform  
Inverse Fourier Transform

# FREQUENCY-DOMAIN REPRESENTATIONS

*Why Frequency domain Representations instead of time-domain representations of signals and systems?*

Speech signals are **non-stationary**. Time-domain provides **amplitude**, **duration** but hides **pitch**, **formants**, **harmonics**.

While the time domain shows *when* a signal changes, the frequency domain shows *what frequencies* make up the signal.



# FOURIER TRANSFORM

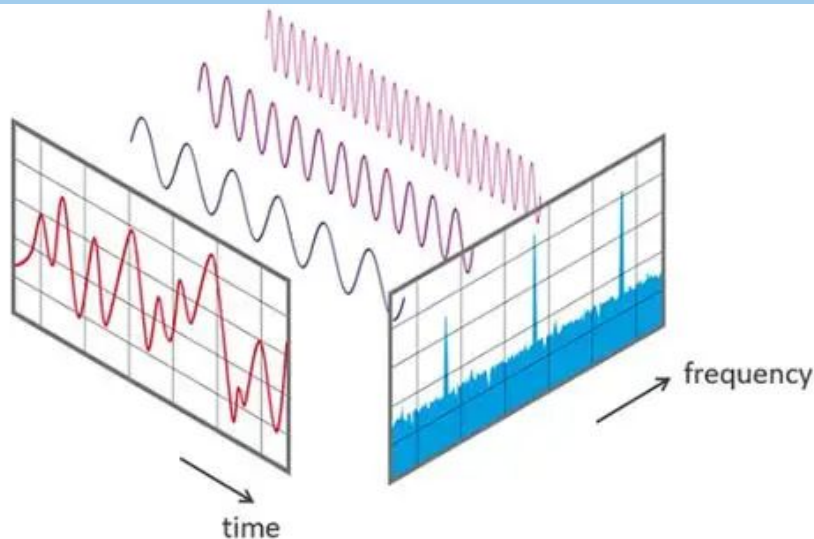
**Fourier Transform** is used for *frequency-domain representations of a signal*

The aim is to understand and analyse the underlying frequency components causing the signal pattern.

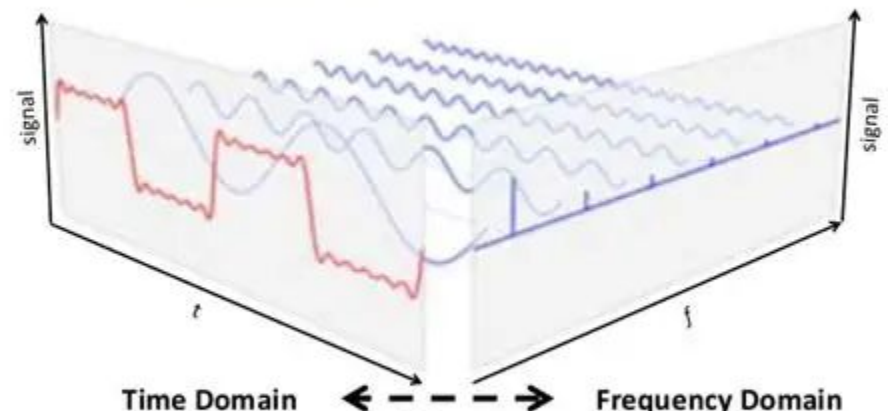
Fourier transform decomposes a signal in terms of sinusoidal (or complex exponential) components with different frequencies, amplitudes, and phases.

The Fourier transform is one of several mathematical tools that is useful in the analysis and design of linear time- invariant (LTI) systems.

Used for **spectral analysis** of speech.



## Fourier Transform - Review



# EXPONENTIAL REPRESENTATION OF SIGNAL

Euler's formula:

$e^{j\theta} = \cos(\theta) + j\sin(\theta) \rightarrow e^{j\theta}$  represents a rotating vector

A sinusoidal signal is represented by

$$s[n] = A \sin(\omega n + \phi) \text{ or } s[n] = A \cos(\omega n + \phi)$$

with  $A$  = amplitude,  $\omega$  = digital angular frequency(rad/sample),

$n$  = sample index.

Euler's formula can express sinusoids as complex exponentials

$$\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$
$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2}$$

# EXPONENTIAL REPRESENTATION OF SIGNAL

Consider the signal:  $3\cos(3\pi t) + 4\cos(2\pi t) + 5\cos(4\pi t)$

Each sinusoid has two complex exponentials.

Here complex exponentials are :  $\frac{3}{2}, 2, \frac{5}{2}$  corresponding to frequencies  $f = \pm 1.5, \pm 1, \pm 2$  Hz.

So the signal can be represented as:

$$x(t) = \frac{3}{2}e^{j3\pi t} + \frac{3}{2}e^{-j3\pi t} + 2e^{j2\pi t} + 2e^{-j2\pi t} + \frac{5}{2}e^{j4\pi t} + \frac{5}{2}e^{-j4\pi t}$$

# FOURIER TRANSFORM

**Fourier transform converts waveform → frequency spectrum**

Fourier transform converts a signal from **time domain to frequency domain**, revealing the frequencies present in a signal and their corresponding amplitudes.

For a continuous-time signal  $x(t)$ , the Fourier Transform is defined as

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt$$

where  $x(t)$  = time-domain signal,

$X(f)$  = frequency-domain representation,

$f$  = frequency (Hz),

$$j = \sqrt{-1}$$

# FOURIER TRANSFORM

Example: Let's consider the following signal:  $x(t) = \cos(2\pi f_0 t)$

$$x(t) = \frac{1}{2}e^{j2\pi f_0 t} + \frac{1}{2}e^{-j2\pi f_0 t}$$

Its Fourier Transform will consist of two impulses:

$$X(f) = \text{Ft}\left\{\frac{1}{2}e^{j2\pi f_0 t} + \frac{1}{2}e^{-j2\pi f_0 t}\right\} = \frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$$

Here,  $\delta(f - f_0)$  indicates that there is energy only at the frequency  $f_0$  and there is no energy at any other frequency.

[a cosine wave contains two frequency components at  $+f_0$  and  $-f_0$ ]

# FOURIER TRANSFORM

Numerical Example: Let's consider the following signal:

$$x(t) = 2\cos(100\pi t) + 3\cos(300\frac{\pi}{4}t)$$

$$x(t) = 2(\frac{1}{2}e^{j2\pi 50t} + \frac{1}{2}e^{-j2\pi 50t}) + 3(\frac{1}{2}e^{j2\pi \frac{300}{8}t} + \frac{1}{2}e^{-j2\pi \frac{300}{8}t})$$

Its Fourier Transform will consist of two impulses

$$\begin{aligned} X(f) &= 2\text{Ft}\left\{\frac{1}{2}e^{j2\pi 50t} + \frac{1}{2}e^{-j2\pi 50t}\right\} + 3F\left\{\frac{1}{2}e^{j2\pi \frac{300}{8}t} + \frac{1}{2}e^{-j2\pi \frac{300}{8}t}\right\} \\ &= \delta(f - 50) + \delta(f + 50) + \frac{3}{2}\delta(f - 37.5) + \frac{3}{2}\delta(f + 37.5) \end{aligned}$$

It indicates that there is energy of magnitude 1 at the frequency 50 and -50. and energy of 1.5 magnitude at the frequency 37.5 and -37.5.

# INVERSE FOURIER-TRANSFORM

Inverse Fourier transform is used to reconstruct the original signal back from the frequency-domain representation.

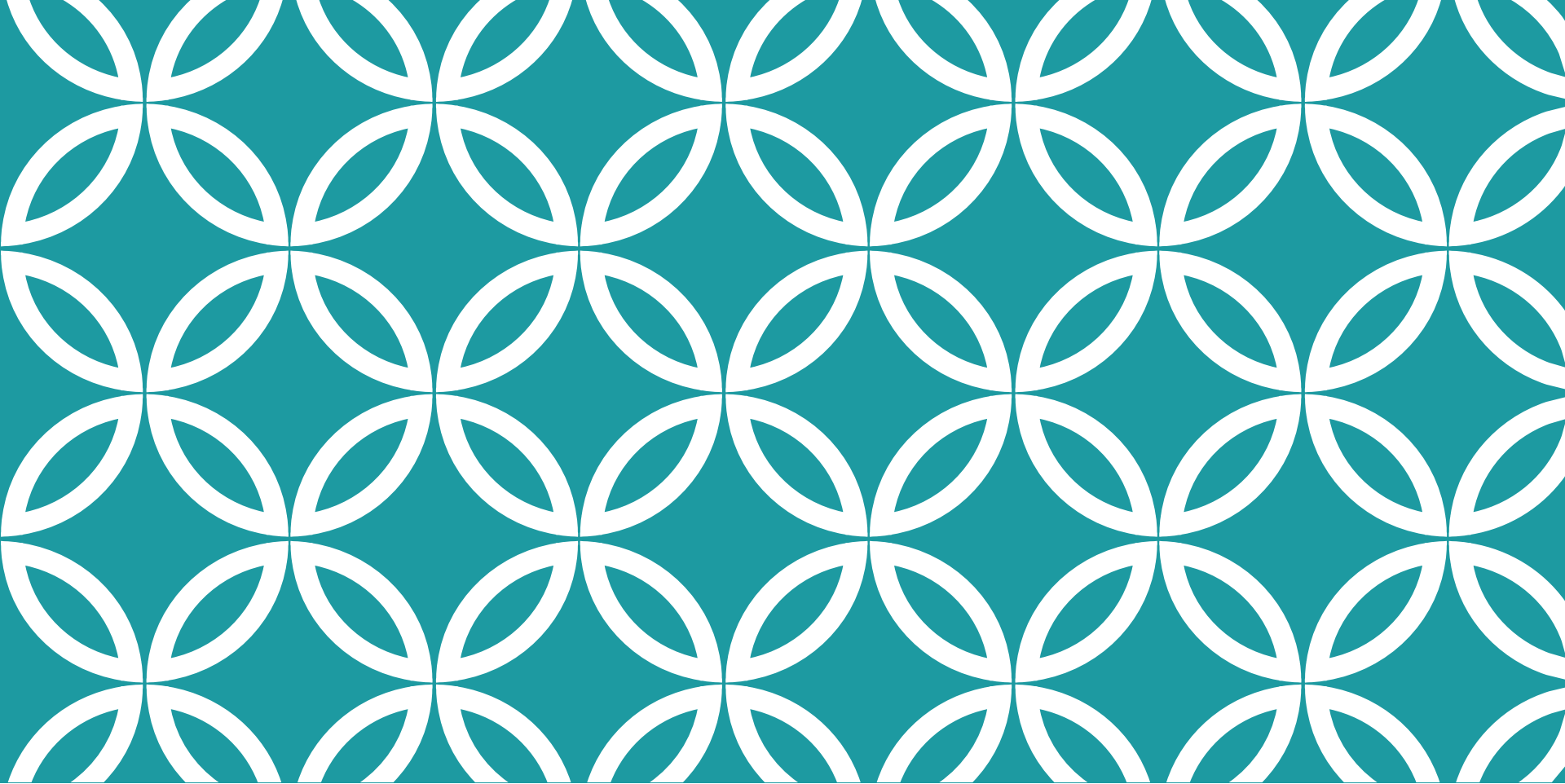
$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

Using angular frequency  $\omega = 2\pi f$ ,

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

# PROPERTIES OF FOURIER TRANSFORM

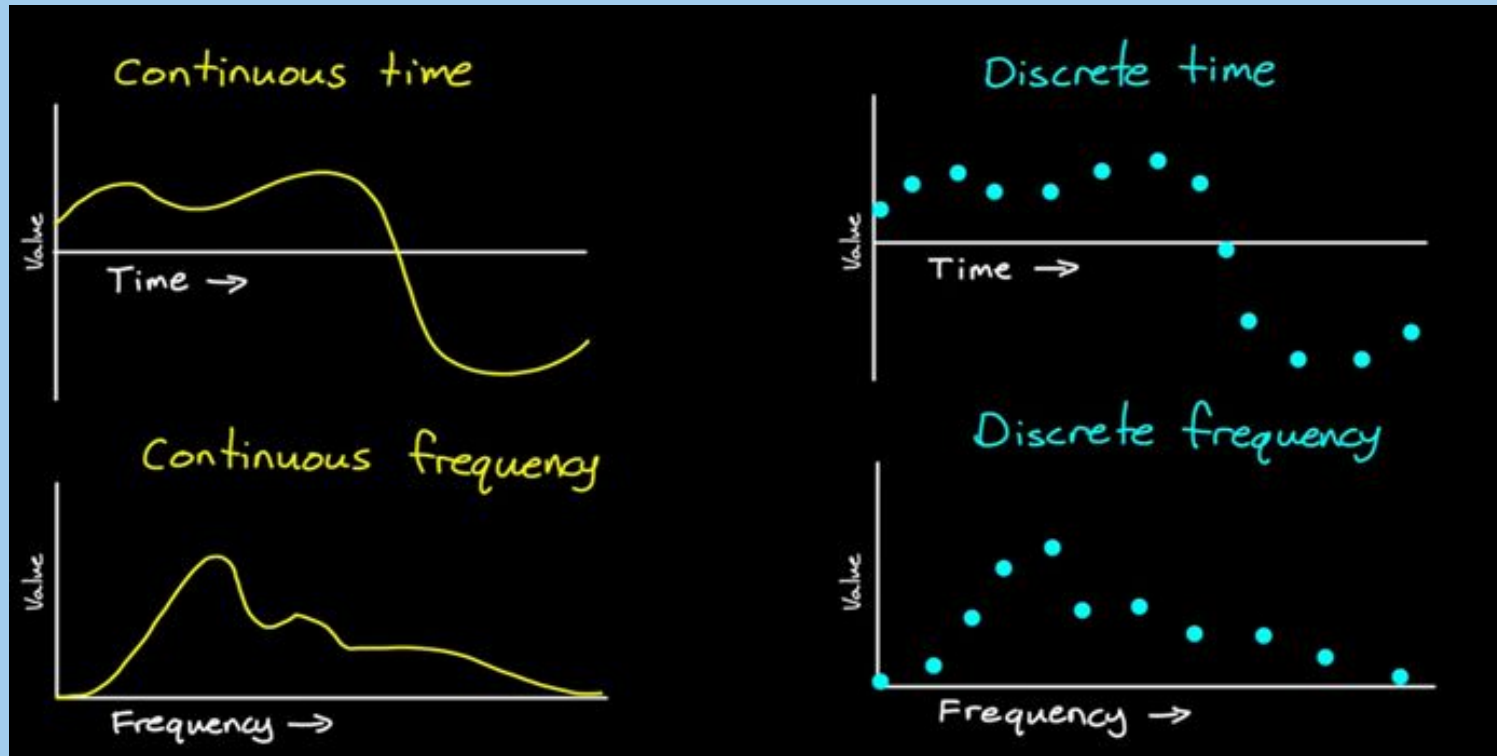
| Property        | Formula |
|-----------------|---------|
| Linearity       |         |
| Time Shift      |         |
| Frequency Shift |         |
| Scaling         |         |
| Convolution     |         |



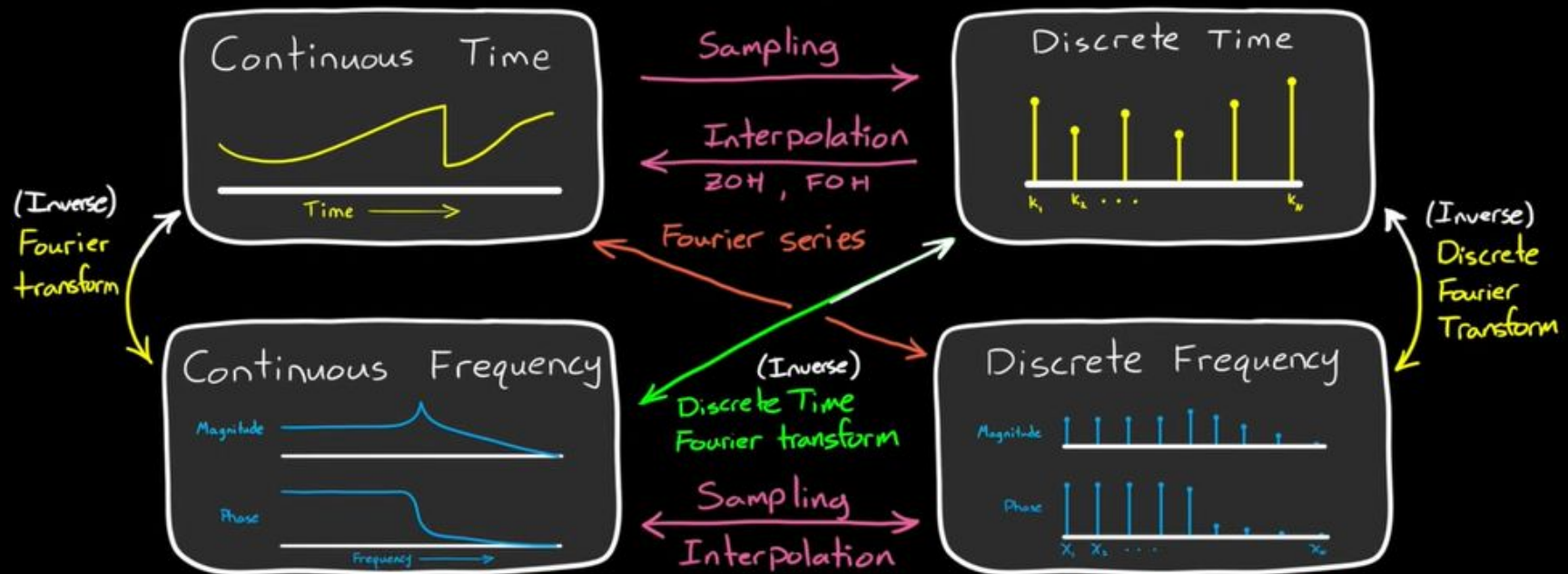
# DISCRETE FOURIER TRANSFORM

Continuous and Discrete  
Frequency  
Discrete-time Fourier  
Transform  
Discrete Fourier  
Transform

# CONTINUOUS AND DISCRETE FREQUENCY



# CONTINUOUS AND DISCRETE FREQUENCY



# DISCRETE-TIME FOURIER TRANSFORM (DTFT)

For a discrete-time signal  $x[n]$ , Fourier transform decomposes the signal into a sum of sinusoidal with different frequencies.

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

$n$  is **discrete** (samples),  $\omega$  (*digital angular frequency*) is **continuous**.

$X(\omega)$  is a **continuous curve**. DTFT shows the spectrum at **every possible frequency**.

In terms of normalized frequency  $f_{norm}$  (continuous),

$$X(f) = \sum_{n=-\infty}^{\infty} x[n]e^{-j2\pi f_{norm}n}$$

DTFT extends the Fourier transform to discrete-time signals, producing a continuous and periodic spectrum.

# DISCRETE-TIME FOURIER TRANSFORM (DTFT)

Inverse DTFT: A large class of sequences can be represented as a linear combination of complex exponentials whose frequencies lie in the range  $[-\pi, \pi]$ .

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

Forward DTFT and inverse DTFT together forms the *discrete-time Fourier transform* pair representation of a sequence.

➤ Why  $-\pi$  to  $\pi$ ?

Sinusoidal signals are periodic. The spectrum repeats every  $2\pi$ .

$$e^{j(\omega+2\pi)n} = e^{j\omega n}$$

Any interval of length  $2\pi$  contains all the unique spectral information

$-\pi$  to  $\pi$  provides a symmetric view of positive and negative frequencies.

# DISCRETE FOURIER TRANSFORM

Discrete Fourier Transform(DFT) samples the DTFT at discrete frequencies for finite-length signals.

DFT is represented as

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$n$  = time index,  $k$  = frequency-bin index,  $N$  = number of samples

DFT is sampled DTFT:  $X[k] = X\left(\omega_k = \frac{2\pi k}{N}\right)$ .  $k$  represents the **DFT frequency bins**.

For each frequency bin  $k$ , the DFT multiplies the signal by a complex exponential. These complex exponentials are called the **basis functions**.

# DISCRETE FOURIER TRANSFORM

**Orthogonality property of DFT basis functions:**

$$\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-m)n} = \begin{cases} N, & m = k \\ 0, & m \neq k \end{cases}$$

If a basis function is compared **with itself** ( $k = m$ ), they are perfectly aligned, so the sum is  $N$ .

If **two different basis functions** are compared ( $k \neq m$ ), their positive and negative contributions cancel over one complete period, so the sum is 0.

Orthogonality guarantees that the DFT cleanly separates frequency components.

# DISCRETE FOURIER TRANSFORM

Example 1: Suppose  $x[n] = 2 \cos\left(\frac{2\pi}{5}n\right) = e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}}$

Consider  $N = 5$ , Digital angular frequencies:  $0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5} = -\frac{4\pi}{5}, \frac{8\pi}{5} = -\frac{2\pi}{5}$

$$x[n] = [2\cos 0, 2\cos \frac{2\pi}{5}, 2\cos \frac{4\pi}{5}, -2\cos \frac{4\pi}{5}, -2\cos \frac{2\pi}{5}] = [2, 0.618, -1.618, -1.618, 0.618]$$

5 bins for 5 samples,  $X[k] = \sum_{n=0}^4 x[n] e^{-j\frac{2\pi}{5}kn}$

The cosine frequency exactly matches **bin 1 and bin 4**.

$$X[0] = 2 + 0.618 - 1.618 - 1.618 + 0.618 = 0$$

$$X[1] = \sum_{n=0}^4 (e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}}) e^{-j2\pi\frac{n}{5}} = \sum_{n=0}^4 1 + \sum_{n=0}^4 e^{-j4\pi\frac{n}{5}} = 5$$

$$X[2] = \sum_{n=0}^4 (e^{j2\pi\frac{n}{5}} + e^{-j2\pi\frac{n}{5}}) e^{-j4\pi\frac{n}{5}} = \sum_{n=0}^4 e^{-j6\pi\frac{n}{5}} + \sum_{n=0}^4 e^{-j2\pi\frac{n}{5}} = 0$$

$$X[3] = 0$$

$$X[4] = 5 \text{ (same as bin 1)}$$

[Note:  $\sum_{n=0}^4 e^{-j4\pi\frac{n}{5}}$ ,  $\sum_{n=0}^4 e^{-j6\pi\frac{n}{5}}$ ,  $\sum_{n=0}^4 e^{-j2\pi\frac{n}{5}}$  value will be zero due to orthogonality property of exponentials]

# DISCRETE FOURIER TRANSFORM

Example 1: Suppose a sine wave with 1 Hz frequency, amplitude 1, sampling frequency: 8 Hz. Apply DFT.

Solution:  $x(t) = \sin(2\pi ft)$

Sampling frequency = 8 Hz, Sampling period =  $1/8$  sec

$$\text{So, } x[n] = 1 \cdot \sin\left(2\pi \cdot 1 \cdot \frac{n}{8}\right) = \sin\left(\frac{\pi}{4}n\right)$$

Frequencies:  $0, \pi/4, (\pi/2), (3\pi/4), (\pi), (5\pi/4), (3\pi/2), (7\pi/4)$

8 bins for 8 samples,  $X[k] = \sum_{n=0}^7 x[n] e^{-j\frac{2\pi}{8}kn}$

Since the sine wave is 1 Hz and sampling is 8 Hz, the normalized frequency is:

$$f_{\text{norm}} = \frac{f}{f_s} = \frac{1}{8}, \text{ so the energy will appear in DFT bin } = f_{\text{norm}} \cdot N = \frac{1}{8} \cdot 8 = 1$$

# DISCRETE FOURIER TRANSFORM

$$\begin{aligned}X[1] &= \sum_{n=0}^7 x[n] e^{-j2\pi \frac{n}{8}} \\&= 0 \cdot e^{-j0} + 0.707 e^{-j\frac{\pi}{4}} + 1 e^{-j\frac{\pi}{2}} + 0.707 e^{-j\frac{3\pi}{4}} + 0 \cdot e^{-j\pi} - 0.707 e^{-j\frac{5\pi}{4}} - 1 \cdot e^{-j\frac{6\pi}{4}} - 0.707 e^{-j\frac{7\pi}{4}} \\&= 0 + (0.5 - j0.5) + (-j) + (-0.5 - j0.5) + 0 + (0.5 - j0.5) + (-j) + (-0.5 - j0.5) \\&\approx -4j\end{aligned}$$

The sine wave's energy is concentrated at **bin 1 (and symmetrically at bin 7)**, confirming the 1 Hz frequency component.

$$X[1] = -4j \rightarrow \text{magnitude } 4, \text{ phase } -90^\circ$$

$$X[7] = +4j \rightarrow \text{magnitude } 4, \text{ phase } +90^\circ$$

# DISCRETE FOURIER TRANSFORM

$$X[1] = \sum_{n=0}^7 x[n] e^{-j2\pi \frac{n}{8}} = \sum_{n=0}^7 \frac{1}{2j} \left( e^{j\pi \frac{n}{4}} - e^{-j\pi \frac{n}{4}} \right) \cdot e^{-j\pi \frac{n}{4}}$$

$$= \frac{1}{2j} \sum_{n=0}^7 e^{j\pi \frac{n}{4}} \cdot e^{-j\pi \frac{n}{4}} - \frac{1}{2j} \sum_{n=0}^7 e^{-j\pi \frac{n}{4}} \cdot e^{-j\pi \frac{n}{4}} = \frac{8}{2j} - 0$$

$$\approx -4j$$

The sine wave's energy is concentrated at **bin 1 (and symmetrically at bin 7)**, confirming the 1 Hz frequency component.

$$X[1] = -4j \rightarrow \text{magnitude 4, phase } -90^\circ$$

$$X[7] = +4j \rightarrow \text{magnitude 4, phase } +90^\circ$$

# DISCRETE FOURIER TRANSFORM

Sine wave: 1Hz, Amplitude = 1

Sampling Frequency: 8 Hz

# samples (N): 8

Fourier Coefficients:

$$X_0 = 0$$

$$X_1 = 0 - 4j$$

$$X_2 = 0$$

$$X_3 = 0$$

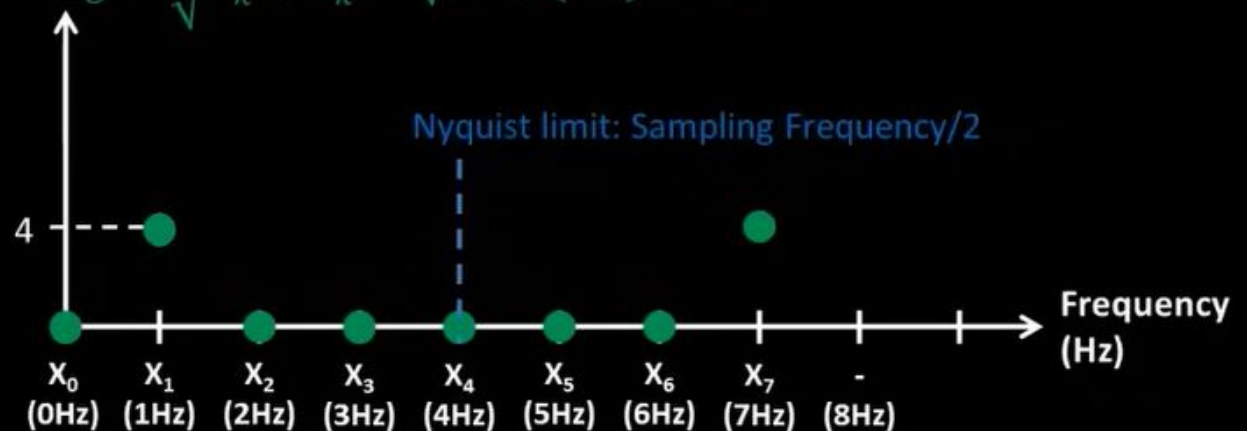
$$X_4 = 0$$

$$X_5 = 0$$

$$X_6 = 0$$

$$X_7 = 0 + 4j$$

$$Mag = \sqrt{A_k^2 + B_k^2} = \sqrt{0^2 + (-4)^2} = 4$$



# DISCRETE FOURIER TRANSFORM

In the previous example, the sine wave's energy is concentrated at **particular bins**.

If the bins align correctly with the sampling frequency, DFT shows clear spikes.

**Clear spikes** at the matching bins (positive and negative frequency components). **Zero elsewhere**, because the orthogonality of exponentials cancels out all other bins.

If the signal frequency does **not** align exactly with a bin, energy spreads across multiple bins (nonzero complex values in several bins). This is called **spectral leakage**.

# COMPLEXITY OF DFT

DFT is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

N number of basis functions.

$N^2$  multiplications and additions. So complexity of DFT is  
 $O(N^2)$

If  $N = 1024$ ,  $\sim 1$  million operations are required.

# EXERCISE

Exercise: Suppose a discrete signal is given as  $x[n] = \{1.5, 0.5, -1.2, 2.5, -2.0\}$ .  
Apply DFT.

Given:  $e^{-j\frac{2\pi}{5}} = 0.31 - j0.95$  and  $e^{-j\frac{4\pi}{5}} = 0.81 - j0.59$

# TWIDDLE FACTOR

Twiddle factor

$$W_N = e^{-\frac{j2\pi}{N}}$$

Twiddle Factors for an 8-point FFT –

$$0^\circ = 1$$

$$-45^\circ = 0.707 - j0.707$$

$$-90^\circ = -j$$

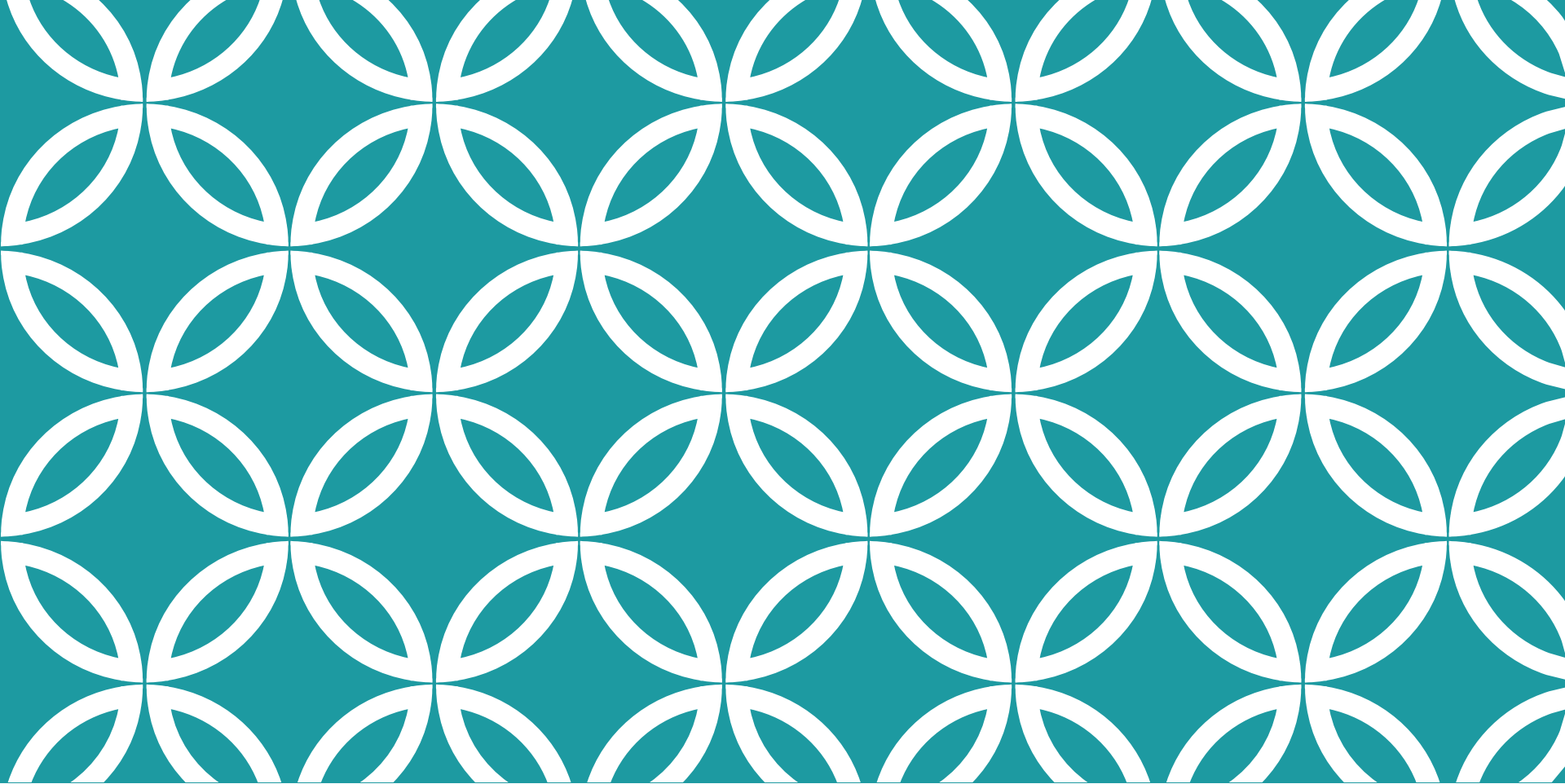
$$-135^\circ = -0.707 - j0.707$$

$$-180^\circ = -1$$

$$-225^\circ = -0.707 + j0.707$$

$$-270^\circ = j$$

$$-315^\circ = 0.707 + j0.707$$



# FAST FOURIER TRANSFORM

Even-Odd Composition  
Twiddle Factor

# FAST FOURIER TRANSFORM

Popularized by **James W. Cooley** and **John W. Tukey** in 1965.

Uses divide and conquer approach to reduce computational complexity of DFT.

Uses Even-Odd decomposition to form smaller subproblems.

By exploiting the symmetry and periodicity of the complex exponential terms, known as **twiddle factors**, the FFT avoids redundant computations and reuses intermediate results.

FFT reduces computational complexity of DFT  $O(N^2)$  to  $O(N \log_2 N)$

*FFT is not a new transform; it is simply a faster algorithm for calculating the DFT.*

# EVEN-ODD DECOMPOSITION

$$F_k = \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-\frac{j2\pi k(2m)}{N}} + \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi k(2m+1)}{N}}$$

**ODD INDEX**

$$+ \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi k\left(m+\frac{1}{2}\right)}{N/2}}$$


$$\sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi km}{N/2} - \frac{j\pi k}{N/2}}$$

# EVEN-ODD DECOMPOSITION

$$F_k = \sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-\frac{j2\pi km}{N/2}} + C_k \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi km}{N/2}}$$

$N = 4$

youtube.com - To exit full screen, swipe down from top of screen or press **Esc**

$$F_k = \sum_{n=0}^{N-1} x_n \cdot e^{-\frac{j2\pi kn}{N}}$$

$$\sum_{m=0}^{N/2-1} x_{2m} \cdot e^{-\frac{j2\pi km}{N/2}}$$

$$e^{-\frac{j2\pi k}{N}} \sum_{m=0}^{N/2-1} x_{2m+1} \cdot e^{-\frac{j2\pi km}{N/2}}$$

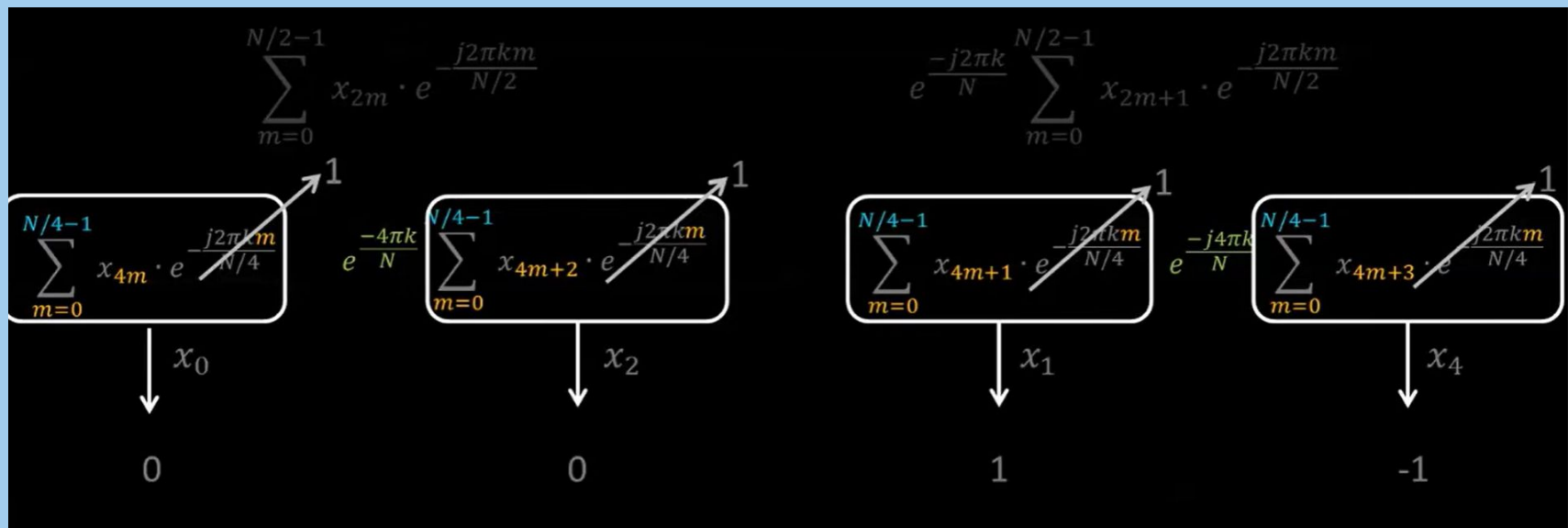
$$\sum_{m=0}^{N/4-1} x_{4m} \cdot e^{-\frac{j2\pi km}{N/4}}$$

$$e^{-\frac{4\pi k}{N}} \sum_{m=0}^{N/4-1} x_{4m+2} \cdot e^{-\frac{j2\pi km}{N/4}}$$

$$\sum_{m=0}^{N/4-1} x_{4m+1} \cdot e^{-\frac{j2\pi km}{N/4}}$$

$$e^{-\frac{j4\pi k}{N}} \sum_{m=0}^{N/4-1} x_{4m+3} \cdot e^{-\frac{j2\pi km}{N/4}}$$

# EVEN-ODD DECOMPOSITION



# EVEN-ODD DECOMPOSITION

$$F_0 = x_0 + x_2 + x_1 + x_3 = \underbrace{0 + 1}_{F_0^e = 1} + \underbrace{0 + (-1)}_{F_0^o = -1} = 0$$

$$\begin{aligned} F_1 &= x_0 + e^{\frac{-j4\pi k}{N}} x_2 + e^{\frac{-j2\pi k}{N}} \left( x_1 + e^{\frac{-j4\pi k}{N}} x_3 \right) \\ &= \underbrace{0 + 1(0)}_{F_1^e = 0} + \underbrace{(-j)(1 + (-1)(-1))}_{F_1^o = 2} = -2j \end{aligned}$$

$$F_2 = F_2^e + e^{\frac{-j2\pi k}{N}} F_2^o$$

$$F_3 = F_3^e + e^{\frac{-j2\pi k}{N}} F_3^o$$

# TWIDDLE FACTOR

Twiddle factor

$$W_N = e^{-\frac{j2\pi}{N}}$$

Twiddle Factors for an 8-point FFT –

$$0^\circ = 1$$

$$-45^\circ = 0.707 - j0.707$$

$$-90^\circ = -j$$

$$-135^\circ = -0.707 - j0.707$$

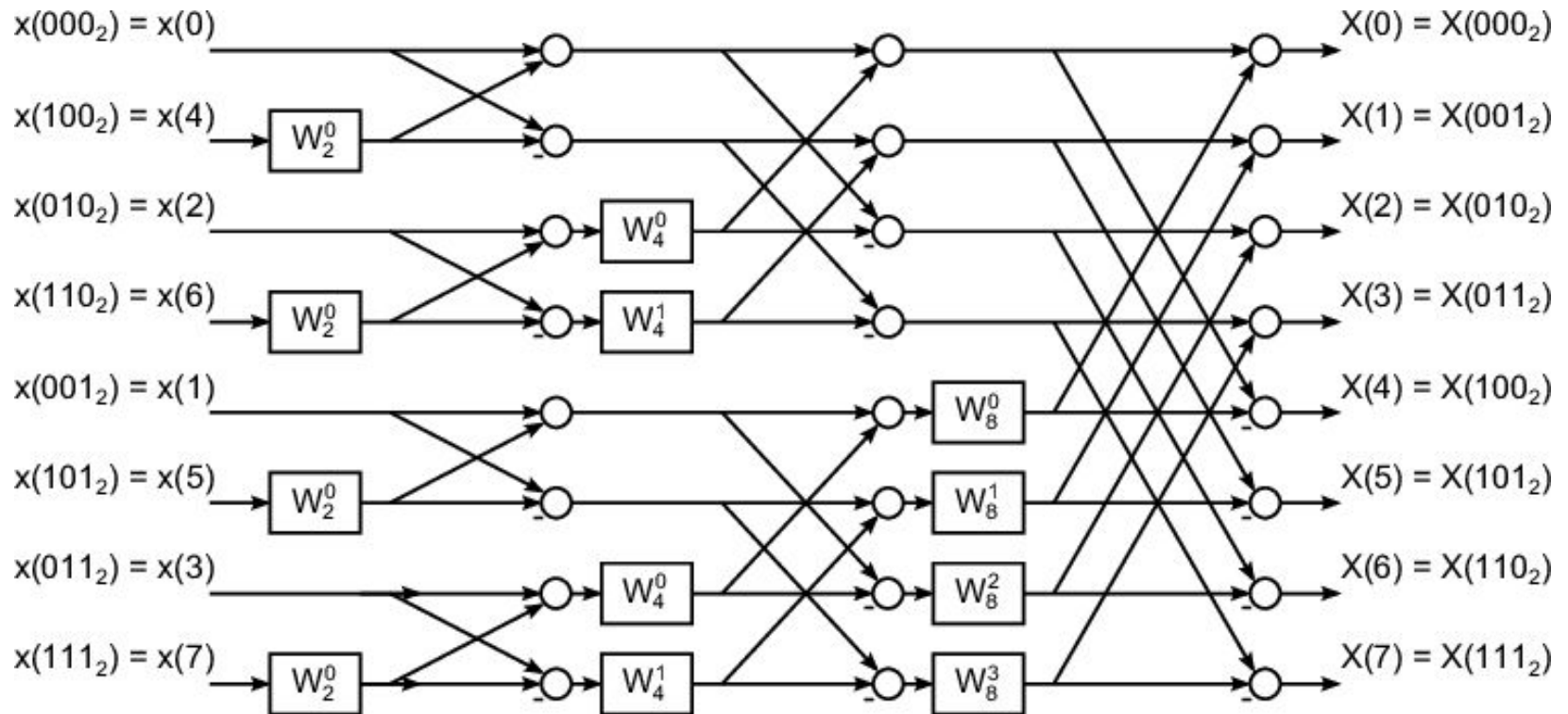
$$-180^\circ = -1$$

$$-225^\circ = -0.707 + j0.707$$

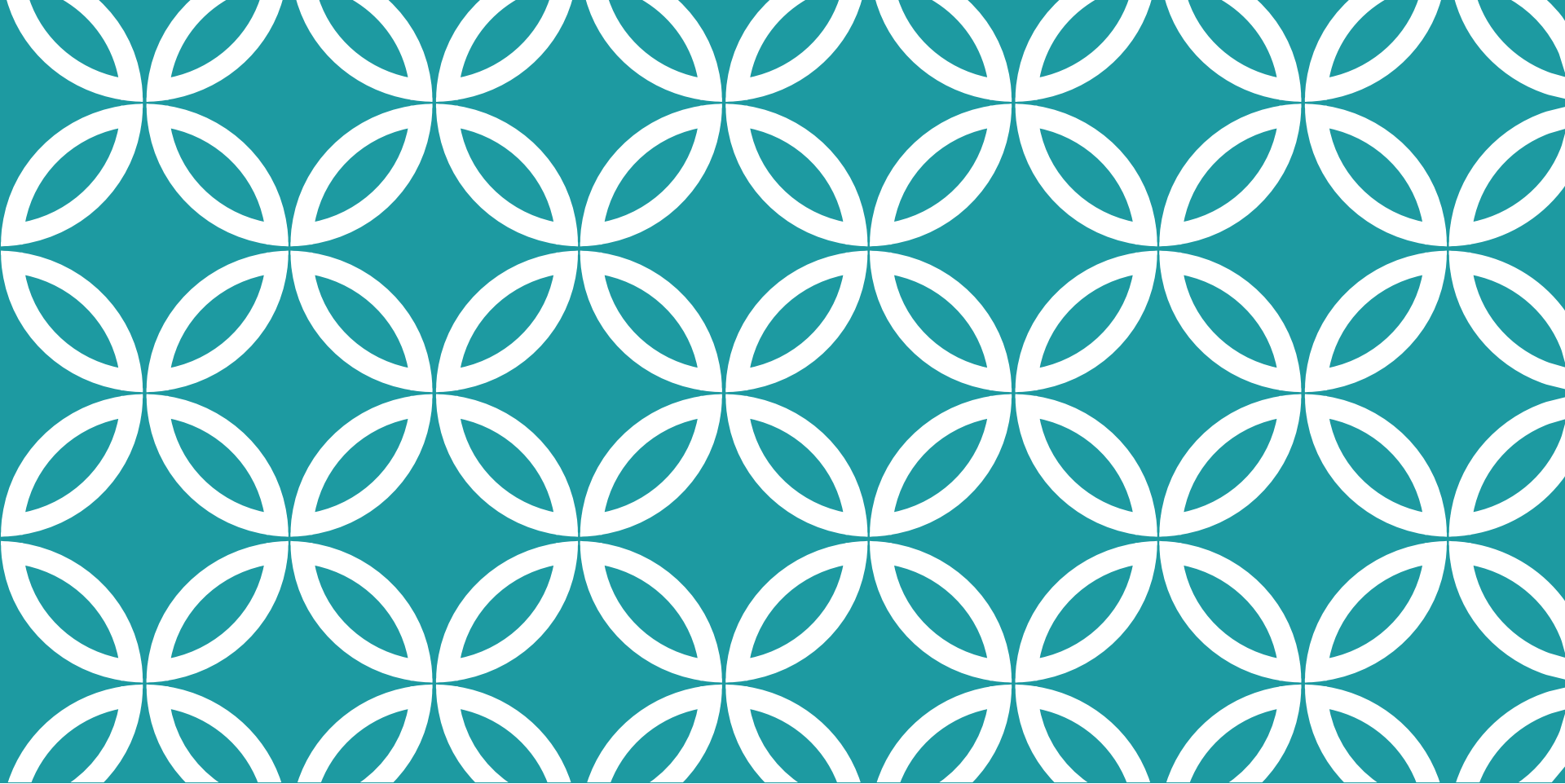
$$-270^\circ = j$$

$$-315^\circ = 0.707 + j0.707$$

# BUTTERFLY NETWORK



Example of an 8-point FFT Butterfly Scheme



# SPECTRAL ANALYSIS

Amplitude Spectrum  
Power Spectrum

# SPECTRAL ANALYSIS

*Spectral Analysis* is a method to study the frequency components of a signal. Using the **Fourier Transform** or **DFT/FFT**, we can analyse which frequencies are present and their relative amplitudes.

Output of DFT/FFT is a set of **complex numbers**:

$$X[k] = \text{Re}[X[k]] + j \cdot \text{Im}[X[k]]$$

These represent both **magnitude** and **phase** of each frequency component.

There are different methods for spectral analysis:

**Amplitude Spectrum** → amplitude vs frequency. Magnitude of FFT coefficients.

**Power Spectrum** → energy vs frequency. Power is the magnitude squared of FFT coefficients (shows energy distribution across frequencies).

**Periodogram** → estimates power spectrum for a finite-length signal.

# SPECTRAL ANALYSIS

